

Junzhou Chen

 junzhouchen.com  [Google Scholar](#)  (+1)9297514535  junzhouchen7@gmail.com

EDUCATION

William & Mary

August 2024 – Present

Ph.D. of Computer Science

Research interest: Vision Language Model, Embodied AI, Video Understanding for VLM, Robotic

Hohai University

August 2020 – June 2024

Bachelor of Intelligent Science and Technology

Average Score: 84/100

As a Robotics and Computer Vision Laboratory member at Hohai University, I have been involved in multiple research projects. Primary research focuses on robotics and artificial intelligence.

PUBLICATIONS

Reasoning Beyond Imitation: Observation-Conditioned Distillation for Video-LLMs

Junzhou Chen, Yuchen Wang, Jindong Wang, Zhou Gang

ICLR 2027 (Under Review)

Fast Enough to Act: Spatio-Temporal Visual Token Merging for Low-Latency Robotic VLMs and VLAs

Junzhou Chen, Jindong Wang, Zhou Gang

IROS 2026

Mitigating In-Transit Vision Noise for Enhanced Vehicle Safety

Yichen Luo, Junzhou Chen, Xinyu Chen, Sidi Lu

ACM Sensys 2025

Edge-Aware Dual Branch Network for Nucleus Instance Segmentation

Junzhou Chen, Yanfu Zhang, Sidi Lu

ACM/IEEE SEC 2024

An Efficient Data Transmission Framework for Connected Vehicle

Yichen Luo, Yongtao Yao, Junzhou Chen, Sidi Lu, Weisong Shi

ACM/IEEE SEC 2024

An Advanced Driving Agent with the Multimodal Large Language Model for Autonomous Vehicles

Junzhou Chen, Sidi Lu

IEEE MOST 2024

A Purified Stacking Ensemble Framework for Cytology Classification

Linyi Qian, Qian Huang, Yulin Chen, Junzhou Chen

MMM 2024

Review of Cervical Cell Segmentation

Qian Huang, Wei Zhang, Yulin Chen, Junzhou Chen

Multimedia Tools and Applications

Novel distributed multi-camera fusion positioning and mapping system and method

Yang Wu, Junzhou Chen, Ziye Wang, Zhengyang Wei, Xuebin Ruan

Patent CN 114943771, 2022

AWARDS AND CERTIFICATES

NSF Student Travel Grant Award @ 2024 ACM/IEEE Symposium on Edge Computing

Best Poster Award @ 2024 Graduate Symposium

International Student Opportunity Scholarship 2024

National College Students' Innovation and Entrepreneurship Training Program 2022

National Excellent Completion

China Internet innovation and Entrepreneurship Competition

Provincial Third Prize

Chinese undergraduate computer design contest 2022

National Second Prize

Chinese undergraduate computer design contest 2021

National Third Prize

Teddy Cup data mining competition

National Third Prize

ROS Junior Engineer

Science and Technology Innovation scholarship

Jiangsu Province Technology Transfer Broker

INTERN & SARTUP EXPERIENCE

The CAR Lab | *Summer Research Intern*

Aug. 2024 - Sep. 2024

- Studied lab-related technologies, including V2X communication and the Blueeyes system.
- Attempted to deploy OpenPilot in the CARLA simulation environment

William and Mary | *Remote Research Assistant*

Aug. 2023 - Aug. 2024

- * Collaborated with Professor Sidi Lu from William & Mary on multimodal large language model for autonomous driving research.
- * Responsible for code development, conducting experiments, and authoring research papers.
- * Produced one research paper currently accepted by MOST 2024

Xuanjia Ling Technology Co., Ltd | *CFO, Co-Founder*

Mar. 2021 – July 2023

- * Spearheaded a student entrepreneurial venture, primarily responsible for commercializing our team's product.
- * Managed data processing, practical and ergonomic design of the product, as well as spearheaded marketing efforts.
- * Temporarily stepped back due to academic commitments, during which I gained extensive experience in product design and development, acquiring a comprehensive understanding of the entire product development process.

RESEARH & PROJECTS

Learning to Reason Across Observations: Video Reasoning Distillation | *Project Leader* May 2026 - Present

- * Developed a black-box knowledge distillation framework to transfer temporal reasoning capabilities from proprietary Video-LLMs to smaller open-source VLMs.
- * Identified the teacher-student observation mismatch in video reasoning, where different temporal sampling processes make teacher-generated reasoning trajectories difficult to directly imitate.
- * Introduced multi-observation reinforcement learning (MO-GRPO) to optimize reasoning across diverse temporal observations, with evaluations spanning multiple in-domain and out-of-domain video reasoning benchmarks.

Research on Efficient Robotic VLMs and VLAs | *Project Leader*

June 2025 - Feb 2026

- * Built a training-free acceleration framework for robotic VLMs and VLAs to reduce visual token redundancy in high-resolution and continuous-input settings.
- * Introduced spatio-temporal token merging with multi-queue matching and positional correction to preserve geometric consistency during aggressive token compression.
- * Achieved 2× faster VLM inference and 8.3× faster VLA control while maintaining task performance on LIBERO and real-world robotic manipulation.

Research on Medical Image Processing Methods | *Segmentation of Overlapping Cell Images* Oct 2022 - Dec 2023

- * The project collaborates with a medical image processing company in Nanjing to develop AI-based pathological analysis of cervical cell slice images. This project has the potential to significantly improve the early diagnosis rate of cancer in patients while greatly reducing the workload of medical professionals.
- * In this project, the focus is on cell segmentation in Barr's cell slice images, with the technical challenge lying in the segmentation of overlapping cells. To address the limitations of traditional network architectures, an innovative HARU-Net network model has been developed, achieving segmentation accuracy at the state-of-the-art level.
- * Code link: <https://github.com/Junzhou-Chen/HARU-Net>

Multi-vehicle Fleet Control Platform | *Vehicle Design and Interface Design*

Nov. 2020 - May 2022

- * This project is a joint development initiative between our team and the Institute of Automation, Chinese Academy of Sciences. In this project, I am responsible for the design of individual vehicles within the vehicle swarm and the interface design.
- * Designing, assembling, and programming the cluster of vehicles, with the vehicles controlled using ESP8266 and connected through a local area network for coordination.
- * The project is based on the Linux system and ROS framework, primarily utilizing C++ and Python for implementation.
- * Project link: <https://www.bilibili.com/video/BV1Ry4y1E7ED/>